

Mode 12 — Square

C-DP4 (C16×C16) → DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → DP8 (4×4) →
add-and-square

Step 1 — C-DP4 (C16×C16)

$$\text{C-DP4 (C16} \times \text{C16)} = D = \sum_{i=0}^3 a_i b_i, \quad a_i = a_i^{re} + j a_i^{im}, \quad b_i = b_i^{re} + j b_i^{im}$$

Step 2 — DP8 (16×16)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{re} - \sum_{i=0}^3 a_i^{im} b_i^{im}}_{\text{DP8 (16} \times \text{16)}}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{im} + \sum_{i=0}^3 a_i^{im} b_i^{re}}_{\text{DP8 (16} \times \text{16)}}$$

Step 3 — DP8 (16×8)

$$\text{Re}(D) = \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{re,hi} - 2^8 \sum_{i=0}^3 a_i^{im} b_i^{im,hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{re,lo} - \sum_{i=0}^3 a_i^{im} b_i^{im,lo}}_{\text{DP8 (16} \times \text{8)}}$$

$$\text{Im}(D) = \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{im,hi} + 2^8 \sum_{i=0}^3 a_i^{im} b_i^{re,hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{im,lo} + \sum_{i=0}^3 a_i^{im} b_i^{re,lo}}_{\text{DP8 (16} \times \text{8)}}$$

Step 4 — DP8 (16×4)

$$\text{Re}(D) = \underbrace{2^{12} \sum_{i=0}^3 a_i^{re} b_i^{re,hh} - 2^{12} \sum_{i=0}^3 a_i^{im} b_i^{im,hh}}_{\text{DP8 (16} \times \text{4)}}$$

$$+ \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{re,hl} - 2^8 \sum_{i=0}^3 a_i^{im} b_i^{im,hl}}_{\text{DP8 (16} \times \text{4)}}$$

$$+ \underbrace{2^4 \sum_{i=0}^3 a_i^{re} b_i^{re,lh} - 2^4 \sum_{i=0}^3 a_i^{im} b_i^{im,lh}}_{\text{DP8 (16} \times \text{4)}}$$

$$+ \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{re,ll} - \sum_{i=0}^3 a_i^{im} b_i^{im,ll}}_{\text{DP8 (16} \times \text{4)}}$$

$$\text{Im}(D) = \underbrace{2^{12} \sum_{i=0}^3 a_i^{re} b_i^{im,hh} + 2^{12} \sum_{i=0}^3 a_i^{im} b_i^{re,hh}}_{\text{DP8 (16} \times \text{4)}}$$

$$+ \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{im,hl} + 2^8 \sum_{i=0}^3 a_i^{im} b_i^{re,hl}}_{\text{DP8 (16} \times \text{4)}}$$

$$\begin{aligned}
& + \underbrace{2^4 \sum_{i=0}^3 a_i^{re} b_i^{im, lh} + 2^4 \sum_{i=0}^3 a_i^{im} b_i^{re, lh}}_{\text{DP8 (16}\times\text{4)}} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{im, ll} + \sum_{i=0}^3 a_i^{im} b_i^{re, ll}}_{\text{DP8 (16}\times\text{4)}}
\end{aligned}$$

Step 5 — DP8 (8×4)

$$\begin{aligned}
\text{Re}(D) = & \underbrace{2^{20} \sum_{i=0}^3 a_i^{re, hi} b_i^{re, hh}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^{20} \sum_{i=0}^3 a_i^{im, hi} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re, lo} b_i^{re, hh}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^{12} \sum_{i=0}^3 a_i^{im, lo} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{16} \sum_{i=0}^3 a_i^{re, hi} b_i^{re, hl}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^{16} \sum_{i=0}^3 a_i^{im, hi} b_i^{im, hl}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re, lo} b_i^{re, hl}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^8 \sum_{i=0}^3 a_i^{im, lo} b_i^{im, hl}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re, hi} b_i^{re, lh}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^{12} \sum_{i=0}^3 a_i^{im, hi} b_i^{im, lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^4 \sum_{i=0}^3 a_i^{re, lo} b_i^{re, lh}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^4 \sum_{i=0}^3 a_i^{im, lo} b_i^{im, lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re, hi} b_i^{re, ll}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^8 \sum_{i=0}^3 a_i^{im, hi} b_i^{im, ll}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re, lo} b_i^{re, ll}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{\sum_{i=0}^3 a_i^{im, lo} b_i^{im, ll}}_{\text{DP8 (8}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
\text{Im}(D) = & \underbrace{2^{20} \sum_{i=0}^3 a_i^{re, hi} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{2^{20} \sum_{i=0}^3 a_i^{im, hi} b_i^{re, hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re, lo} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{2^{12} \sum_{i=0}^3 a_i^{im, lo} b_i^{re, hh}}_{\text{DP8 (8}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^{16} \underbrace{\sum_{i=0}^3 a_i^{re,hi} b_i^{im,hl}}_{\text{DP8 } (8 \times 4)} + 2^{16} \underbrace{\sum_{i=0}^3 a_i^{im,hi} b_i^{re,hl}}_{\text{DP8 } (8 \times 4)} \\
& + 2^8 \underbrace{\sum_{i=0}^3 a_i^{re,lo} b_i^{im,hl}}_{\text{DP8 } (8 \times 4)} + 2^8 \underbrace{\sum_{i=0}^3 a_i^{im,lo} b_i^{re,hl}}_{\text{DP8 } (8 \times 4)} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_i^{re,hi} b_i^{im,lh}}_{\text{DP8 } (8 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^3 a_i^{im,hi} b_i^{re,lh}}_{\text{DP8 } (8 \times 4)} \\
& + 2^4 \underbrace{\sum_{i=0}^3 a_i^{re,lo} b_i^{im,lh}}_{\text{DP8 } (8 \times 4)} + 2^4 \underbrace{\sum_{i=0}^3 a_i^{im,lo} b_i^{re,lh}}_{\text{DP8 } (8 \times 4)} \\
& + 2^8 \underbrace{\sum_{i=0}^3 a_i^{re,hi} b_i^{im,ll}}_{\text{DP8 } (8 \times 4)} + 2^8 \underbrace{\sum_{i=0}^3 a_i^{im,hi} b_i^{re,ll}}_{\text{DP8 } (8 \times 4)} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re,lo} b_i^{im,ll}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=0}^3 a_i^{im,lo} b_i^{re,ll}}_{\text{DP8 } (8 \times 4)}
\end{aligned}$$

Step 6 — DP8 (4×4)

$$\begin{aligned}
\text{Re}(D) = & 2^{24} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{re,hh}}_{\text{DP8 } (4 \times 4)} + 2^{20} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{re,hh}}_{\text{DP8 } (4 \times 4)} \\
& - 2^{24} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{im,hh}}_{\text{DP8 } (4 \times 4)} - 2^{20} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{im,hh}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{re,hh}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{re,hh}}_{\text{DP8 } (4 \times 4)} \\
& - 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{im,hh}}_{\text{DP8 } (4 \times 4)} - 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{im,hh}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{20} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{re,hl}}_{\text{DP8 } (4 \times 4)} + 2^{16} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{re,hl}}_{\text{DP8 } (4 \times 4)} \\
& - 2^{20} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{im,hl}}_{\text{DP8 } (4 \times 4)} - 2^{16} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{im,hl}}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

$$\begin{aligned}
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}} \\
& - 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{im,hl}}_{\text{DP8 (4}\times\text{4)}} - 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{im,hl}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{re,lh}}_{\text{DP8 (4}\times\text{4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{re,lh}}_{\text{DP8 (4}\times\text{4)}} \\
& - 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{im,lh}}_{\text{DP8 (4}\times\text{4)}} - 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{im,lh}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{re,lh}}_{\text{DP8 (4}\times\text{4)}} + 2^4 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{re,lh}}_{\text{DP8 (4}\times\text{4)}} \\
& - 2^8 \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{im,lh}}_{\text{DP8 (4}\times\text{4)}} - 2^4 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{im,lh}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{re,ll}}_{\text{DP8 (4}\times\text{4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{re,ll}}_{\text{DP8 (4}\times\text{4)}} \\
& - 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{im,ll}}_{\text{DP8 (4}\times\text{4)}} - 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{im,ll}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{re,ll}}_{\text{DP8 (4}\times\text{4)}} + \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{re,ll}}_{\text{DP8 (4}\times\text{4)}} \\
& - 2^4 \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{im,ll}}_{\text{DP8 (4}\times\text{4)}} - \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{im,ll}}_{\text{DP8 (4}\times\text{4)}} \\
\text{Im}(D) = & 2^{24} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{im,hh}}_{\text{DP8 (4}\times\text{4)}} + 2^{20} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{im,hh}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{24} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{re,hh}}_{\text{DP8 (4}\times\text{4)}} + 2^{20} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{re,hh}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{im,hh}}_{\text{DP8 (4}\times\text{4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{im,hh}}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{re,hh}}_{\text{DP8 (4}\times\text{4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{re,hh}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{20} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{im,hl}}_{\text{DP8 (4}\times\text{4)}} + 2^{16} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{im,hl}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{20} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}} + 2^{16} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{im,hl}}_{\text{DP8 (4}\times\text{4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{im,hl}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{im,lh}}_{\text{DP8 (4}\times\text{4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{im,lh}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{re,lh}}_{\text{DP8 (4}\times\text{4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{re,lh}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{im,lh}}_{\text{DP8 (4}\times\text{4)}} + 2^4 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{im,lh}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{re,lh}}_{\text{DP8 (4}\times\text{4)}} + 2^4 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{re,lh}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{im,ll}}_{\text{DP8 (4}\times\text{4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{im,ll}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{re,ll}}_{\text{DP8 (4}\times\text{4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{re,ll}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{im,ll}}_{\text{DP8 (4}\times\text{4)}} + \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{im,ll}}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{re,ll}}_{\text{DP8 (4}\times\text{4)}} + \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{re,ll}}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

Step 7 — Add-and-square (substitute each DP8 $(4 \times 4) = \frac{1}{2} [\sum (x_i + b_i)^2 - \sum x_i^2 - \sum b_i^2]$)

$$\begin{aligned}
\text{Re}(D) = & \underbrace{2^{24} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{20} \frac{1}{2} \left(\sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{24} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{im,hi} - b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{20} \frac{1}{2} \left(\sum_{i=0}^3 (a_{L,i}^{im,hi} - b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{16} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{12} \frac{1}{2} \left(\sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{16} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{im,lo} - b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{12} \frac{1}{2} \left(\sum_{i=0}^3 (a_{L,i}^{im,lo} - b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{20} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,hl})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{re,hl})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{16} \frac{1}{2} \left(\sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{re,hl})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{re,hl})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{20} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{im,hi} - b_i^{im,hl})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{im,hl})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \underbrace{2^{16} \frac{1}{2} \left(\sum_{i=0}^3 (a_{L,i}^{im,hi} - b_i^{im,hl})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{im,hl})^2 \right)}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

$$\begin{aligned}
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{re,hl})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{re,hl})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{re,hl})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{re,hl})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,lo} - b_i^{im,hl})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{im,hl})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,lo} - b_i^{im,hl})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{im,hl})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,lh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{re,lh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{re,lh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{re,lh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} - b_i^{im,lh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{im,lh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,hi} - b_i^{im,lh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{im,lh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{re,lh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{re,lh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{re,lh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{re,lh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,lo} - b_i^{im,lh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{im,lh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,lo} - b_i^{im,lh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{im,lh})^2 \right)}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

$$\begin{aligned}
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,ll})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{re,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{re,ll})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{re,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} - b_i^{im,ll})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{im,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,hi} - b_i^{im,ll})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{im,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{re,ll})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{re,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{re,ll})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{re,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,lo} - b_i^{im,ll})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{im,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,lo} - b_i^{im,ll})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{im,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
\text{Im}(D) = & 2^{24} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{24} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,hi} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,lo} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,lo} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,hl})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{im,hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{im,hl})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{im,hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,hl})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{re,hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,hi} + b_i^{re,hl})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{re,hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{im,hl})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{im,hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{im,hl})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{im,hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,lo} + b_i^{re,hl})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{re,hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,lo} + b_i^{re,hl})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{re,hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,lh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{im,lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{im,lh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{im,lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,lh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{re,lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,hi} + b_i^{re,lh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{re,lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{im,lh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{im,lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{im,lh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{im,lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,lo} + b_i^{re,lh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{re,lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,lo} + b_i^{re,lh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{re,lh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,ll})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{im,ll})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{im,ll})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{im,ll})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,ll})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{re,ll})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,hi} + b_i^{re,ll})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{re,ll})^2 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{im,ll})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{im,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{im,ll})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 - \sum_{i=0}^3 (b_i^{im,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,lo} + b_i^{re,ll})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{re,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{L,i}^{im,lo} + b_i^{re,ll})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 - \sum_{i=0}^3 (b_i^{re,ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

Step 8 — All components

Re(D):

$$\begin{aligned}
\text{PE} = & 2^{24} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,hh})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{re,hh})^2 + 2^{24} \sum_{i=0}^3 (a_{H,i}^{im,hi} - b_i^{im,hh})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{im,hi} - b_i^{im,hh})^2 \\
& + 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{re,hh})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{re,hh})^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,lo} - b_i^{im,hh})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,lo} - b_i^{im,hh})^2 \\
& + 2^{20} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,hl})^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{re,hl})^2 + 2^{20} \sum_{i=0}^3 (a_{H,i}^{im,hi} - b_i^{im,hl})^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{im,hi} - b_i^{im,hl})^2 \\
& + 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{re,hl})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{re,hl})^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,lo} - b_i^{im,hl})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,lo} - b_i^{im,hl})^2 \\
& + 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,lh})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{re,lh})^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,hi} - b_i^{im,lh})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,hi} - b_i^{im,lh})^2 \\
& + 2^8 \sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{re,lh})^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{re,lh})^2 + 2^8 \sum_{i=0}^3 (a_{H,i}^{im,lo} - b_i^{im,lh})^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{im,lo} - b_i^{im,lh})^2 \\
& + 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,ll})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{re,ll})^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,hi} - b_i^{im,ll})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,hi} - b_i^{im,ll})^2 \\
& + 2^4 \sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{re,ll})^2 + \sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{re,ll})^2 + 2^4 \sum_{i=0}^3 (a_{H,i}^{im,lo} - b_i^{im,ll})^2 + \sum_{i=0}^3 (a_{L,i}^{im,lo} - b_i^{im,ll})^2 \\
\alpha = & 2^{24} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 + 2^{24} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 \\
& + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 + 2^{20} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 + 2^{20} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 \\
& + 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 \\
& + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 + 2^8 \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 + 2^8 \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 \\
& + 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 + 2^4 \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 + \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 \\
& + 2^4 \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 + \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 \\
\beta = & 2^{24} \sum_{i=0}^3 (b_i^{re,hh})^2 + 2^{20} \sum_{i=0}^3 (b_i^{re,hh})^2 + 2^{24} \sum_{i=0}^3 (b_i^{im,hh})^2 + 2^{20} \sum_{i=0}^3 (b_i^{im,hh})^2 + 2^{16} \sum_{i=0}^3 (b_i^{re,hh})^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,hh})^2 \\
& + 2^{16} \sum_{i=0}^3 (b_i^{im,hh})^2 + 2^{12} \sum_{i=0}^3 (b_i^{im,hh})^2 + 2^{20} \sum_{i=0}^3 (b_i^{re,hl})^2 + 2^{16} \sum_{i=0}^3 (b_i^{re,hl})^2 + 2^{20} \sum_{i=0}^3 (b_i^{im,hl})^2 + 2^{16} \sum_{i=0}^3 (b_i^{im,hl})^2 \\
& + 2^{12} \sum_{i=0}^3 (b_i^{re,hl})^2 + 2^8 \sum_{i=0}^3 (b_i^{re,hl})^2 + 2^{12} \sum_{i=0}^3 (b_i^{im,hl})^2 + 2^8 \sum_{i=0}^3 (b_i^{im,hl})^2 + 2^{16} \sum_{i=0}^3 (b_i^{re,lh})^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,lh})^2 \\
& + 2^{16} \sum_{i=0}^3 (b_i^{im,lh})^2 + 2^{12} \sum_{i=0}^3 (b_i^{im,lh})^2 + 2^8 \sum_{i=0}^3 (b_i^{re,lh})^2 + 2^4 \sum_{i=0}^3 (b_i^{re,lh})^2 + 2^8 \sum_{i=0}^3 (b_i^{im,lh})^2 + 2^4 \sum_{i=0}^3 (b_i^{im,lh})^2 \\
& + 2^{12} \sum_{i=0}^3 (b_i^{re,ll})^2 + 2^8 \sum_{i=0}^3 (b_i^{re,ll})^2 + 2^{12} \sum_{i=0}^3 (b_i^{im,ll})^2 + 2^8 \sum_{i=0}^3 (b_i^{im,ll})^2 + 2^4 \sum_{i=0}^3 (b_i^{re,ll})^2 + \sum_{i=0}^3 (b_i^{re,ll})^2 \\
& + 2^4 \sum_{i=0}^3 (b_i^{im,ll})^2 + \sum_{i=0}^3 (b_i^{im,ll})^2
\end{aligned}$$

$\text{Im}(D)$:

$$\begin{aligned}
\text{PE} = & 2^{24} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,hh})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{im,hh})^2 + 2^{24} \sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,hh})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{im,hi} + b_i^{re,hh})^2 \\
& + 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{im,hh})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{im,hh})^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,lo} + b_i^{re,hh})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,lo} + b_i^{re,hh})^2 \\
& + 2^{20} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,hl})^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{im,hl})^2 + 2^{20} \sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,hl})^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{im,hi} + b_i^{re,hl})^2 \\
& + 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{im,hl})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{im,hl})^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,lo} + b_i^{re,hl})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,lo} + b_i^{re,hl})^2 \\
& + 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,lh})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{im,lh})^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,lh})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,hi} + b_i^{re,lh})^2 \\
& + 2^8 \sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{im,lh})^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{im,lh})^2 + 2^8 \sum_{i=0}^3 (a_{H,i}^{im,lo} + b_i^{re,lh})^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{im,lo} + b_i^{re,lh})^2 \\
& + 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,ll})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,hi} + b_i^{im,ll})^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,ll})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,hi} + b_i^{re,ll})^2 \\
& + 2^4 \sum_{i=0}^3 (a_{H,i}^{re,lo} + b_i^{im,ll})^2 + \sum_{i=0}^3 (a_{L,i}^{re,lo} + b_i^{im,ll})^2 + 2^4 \sum_{i=0}^3 (a_{H,i}^{im,lo} + b_i^{re,ll})^2 + \sum_{i=0}^3 (a_{L,i}^{im,lo} + b_i^{re,ll})^2 \\
\alpha = & 2^{24} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 + 2^{24} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 \\
& + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 + 2^{20} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 + 2^{20} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 \\
& + 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 \\
& + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 + 2^8 \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 + 2^8 \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 \\
& + 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,hi})^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,hi})^2 + 2^4 \sum_{i=0}^3 (a_{H,i}^{re,lo})^2 + \sum_{i=0}^3 (a_{L,i}^{re,lo})^2 \\
& + 2^4 \sum_{i=0}^3 (a_{H,i}^{im,lo})^2 + \sum_{i=0}^3 (a_{L,i}^{im,lo})^2 \\
\beta = & 2^{24} \sum_{i=0}^3 (b_i^{im,hh})^2 + 2^{20} \sum_{i=0}^3 (b_i^{im,hh})^2 + 2^{24} \sum_{i=0}^3 (b_i^{re,hh})^2 + 2^{20} \sum_{i=0}^3 (b_i^{re,hh})^2 + 2^{16} \sum_{i=0}^3 (b_i^{im,hh})^2 + 2^{12} \sum_{i=0}^3 (b_i^{im,hh})^2 \\
& + 2^{16} \sum_{i=0}^3 (b_i^{re,hh})^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,hh})^2 + 2^{20} \sum_{i=0}^3 (b_i^{im,hl})^2 + 2^{16} \sum_{i=0}^3 (b_i^{im,hl})^2 + 2^{20} \sum_{i=0}^3 (b_i^{re,hl})^2 + 2^{16} \sum_{i=0}^3 (b_i^{re,hl})^2 \\
& + 2^{12} \sum_{i=0}^3 (b_i^{im,hl})^2 + 2^8 \sum_{i=0}^3 (b_i^{im,hl})^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,hl})^2 + 2^8 \sum_{i=0}^3 (b_i^{re,hl})^2 + 2^{16} \sum_{i=0}^3 (b_i^{im,lh})^2 + 2^{12} \sum_{i=0}^3 (b_i^{im,lh})^2 \\
& + 2^{16} \sum_{i=0}^3 (b_i^{re,lh})^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,lh})^2 + 2^8 \sum_{i=0}^3 (b_i^{im,lh})^2 + 2^4 \sum_{i=0}^3 (b_i^{im,lh})^2 + 2^8 \sum_{i=0}^3 (b_i^{re,lh})^2 + 2^4 \sum_{i=0}^3 (b_i^{re,lh})^2 \\
& + 2^{12} \sum_{i=0}^3 (b_i^{im,ll})^2 + 2^8 \sum_{i=0}^3 (b_i^{im,ll})^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,ll})^2 + 2^8 \sum_{i=0}^3 (b_i^{re,ll})^2 + 2^4 \sum_{i=0}^3 (b_i^{im,ll})^2 + \sum_{i=0}^3 (b_i^{im,ll})^2 \\
& + 2^4 \sum_{i=0}^3 (b_i^{re,ll})^2 + \sum_{i=0}^3 (b_i^{re,ll})^2
\end{aligned}$$